

THE MASTER COMPENDIUM

Electrical Machines II - Induction Machine Mastery

Phase 5

Power, Torque & Performance Dynamics

This phase converts the equivalent circuit into torque, output power, efficiency and performance curves. The aim is not to memorize equations, but to see how each branch of the circuit becomes a physical power path.

Focus: air-gap power, rotor copper loss, developed mechanical power, torque equation, maximum torque, maximum power, algebra ratios, and torque-speed curve.

Prepared for semester examination revision.

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1 Where Phase 5 Fits in the Roadmap

Phase-to-phase continuity

In Phase 4, the machine was translated into a per-phase equivalent circuit. Phase 5 uses that circuit to answer the most important practical question:

For a given voltage and slip, how much torque and output power will the motor produce?

The rotor branch is the engine room. The resistance R'_2/s receives air-gap power. The part R'_2 becomes rotor copper loss, while the part $R'_2(1-s)/s$ represents developed mechanical power.

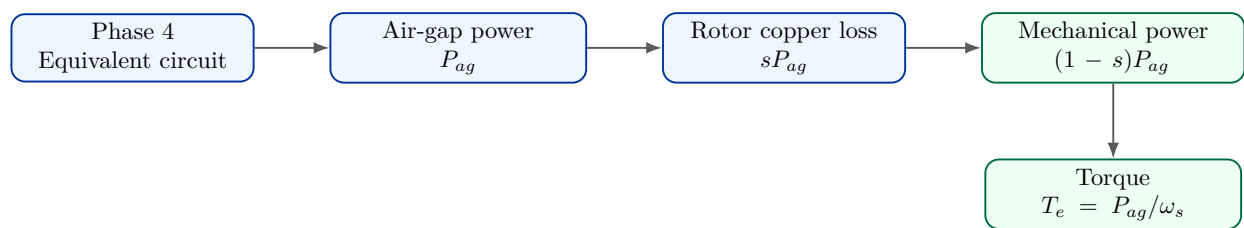


Figure 1: Mental flow from Phase 4 to Phase 5. First circuit, then power, then torque.

What you must be able to do after Phase 5

1. Derive the torque equation from the equivalent circuit.
2. Explain why torque is proportional to square of voltage.
3. Derive maximum torque slip and prove maximum torque is independent of rotor resistance.
4. Draw the correct torque-speed curve with starting torque, full-load point and pull-out torque.
5. Solve numericals by following power flow: input $\rightarrow$ stator loss $\rightarrow$ air gap $\rightarrow$ rotor loss $\rightarrow$ mechanical output.

2 Correctly Joined Circuits for Phase 5

The most common mistake is to draw the rotor load resistance as if it is floating. It must always be part of a closed branch. The bottom return line is as important as the top line. Without a continuous return path, the circuit is not electrically meaningful.

2.1 Complete per-phase equivalent circuit

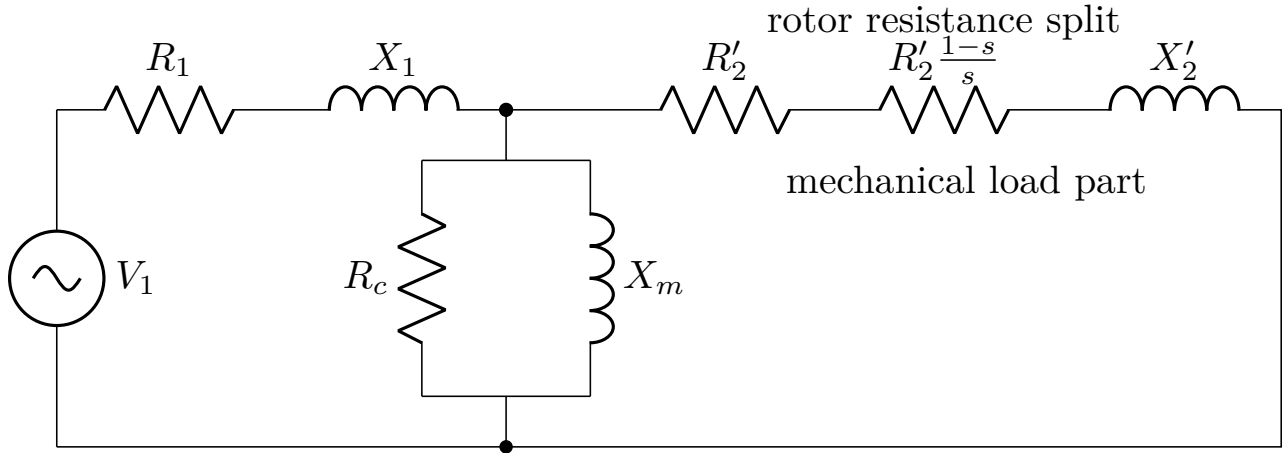


Figure 2: Correct complete equivalent circuit. The rotor branch is closed through the bottom return line. The mechanical load resistance is not floating.

$$\frac{R_2'}{s} = R_2' + R_2' \left(\frac{1-s}{s} \right)$$

Part	Physical meaning
R_2'	Rotor copper loss resistance. Power in this part is lost as heat in rotor conductors.
$R_2'(1-s)/s$	Mechanical power conversion resistance. Power in this part is converted into developed mechanical power.
R_2'/s	Total effective rotor resistance seen in the slip-frequency corrected circuit.

2.2 Approximate circuit used for torque derivation

For exam derivations, the shunt branch is often shifted or neglected in the final torque expression. Then the motor behaves like a voltage V feeding a series impedance $R_1 + R_2'/s + j(X_1 + X_2')$.

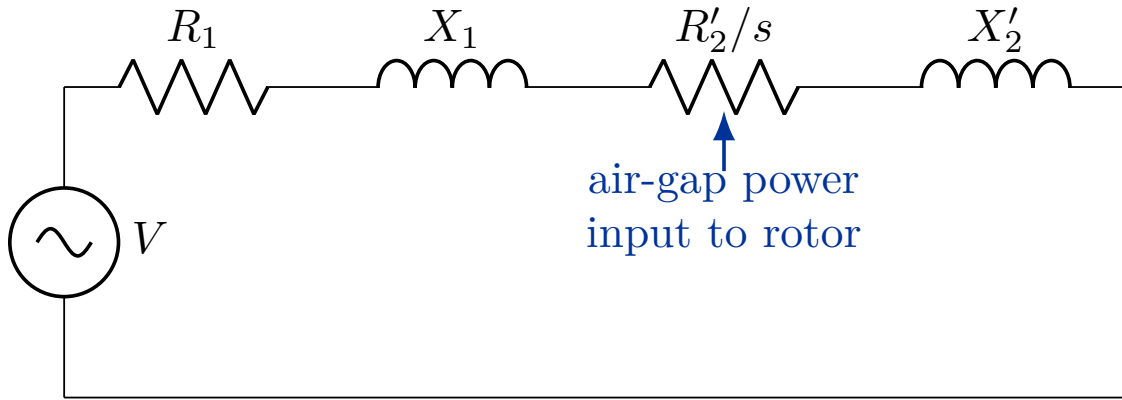


Figure 3: Simplified series circuit. This is the cleanest circuit for deriving the basic torque equation.

2.3 Thevenin equivalent circuit for accurate torque equation

When the magnetizing branch is retained properly, the stator side is replaced by a Thevenin voltage and impedance as seen from the air-gap terminals.

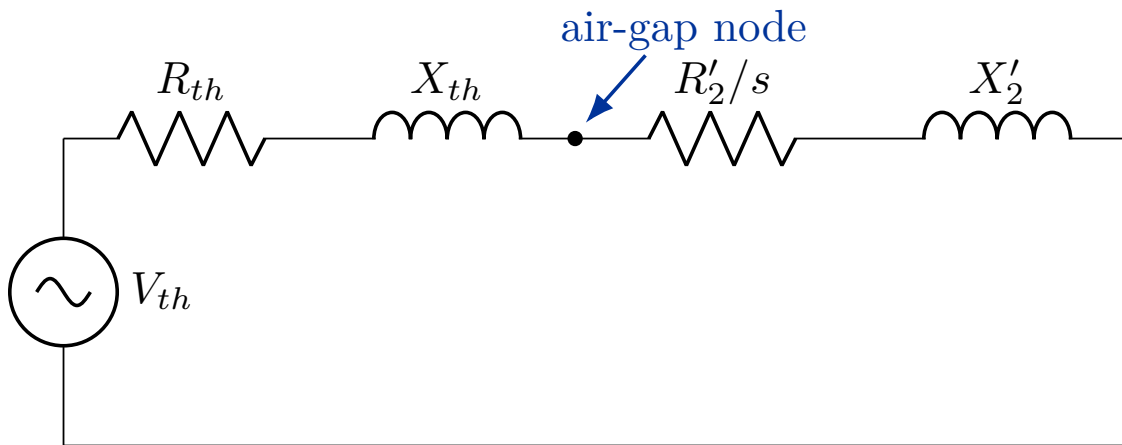


Figure 4: Thevenin equivalent circuit. This gives the more accurate torque expression.

Circuit joining checklist

1. Every element must have two connected ends.
2. The rotor branch must return to the same bottom reference line.
3. The mechanical resistance $R_2'(1 - s)/s$ is in series with R_2' , not in parallel with the supply.
4. The current through R_2'/s is rotor current referred to stator, I_2' .

3 Power Flow: The Backbone of All Numericals

The one-line story

Electrical input power first suffers stator losses. The remaining power crosses the air gap. In the rotor, a fraction s becomes rotor copper loss and the fraction $1 - s$ becomes developed mechanical power.

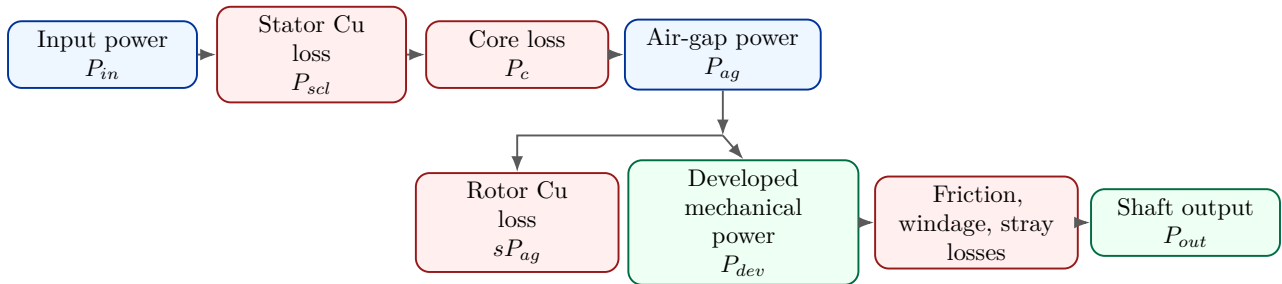


Figure 5: Induction motor power-flow diagram. In exams, draw this before starting any numerical.

$$P_{in} = \sqrt{3}V_L I_L \cos \phi,$$

$$P_{scl} = 3I_1^2 R_1,$$

$$P_{ag} = P_{in} - P_{scl} - P_c,$$

$$P_{rcl} = sP_{ag},$$

$$P_{dev} = P_{ag} - P_{rcl} = (1 - s)P_{ag},$$

$$P_{out} = P_{dev} - P_{rotational\ losses}.$$

Fast slip trick from power flow

If air-gap power and rotor copper loss are known:

$$s = \frac{P_{rcl}}{P_{ag}}$$

This is one of the fastest ways to find slip in numerical problems.

4 Air-gap Power and the Meaning of R'_2/s

From the rotor branch of the referred equivalent circuit,

$$P_{ag} = 3(I'_2)^2 \left(\frac{R'_2}{s} \right).$$

This looks like ordinary three-phase copper power. But because the rotor frequency has been converted to stator frequency, the effective resistance becomes R'_2/s .

Splitting air-gap power into rotor loss and mechanical power

Start from:

$$P_{ag} = 3(I'_2)^2 \left(\frac{R'_2}{s} \right)$$

Split the resistance:

$$\frac{R'_2}{s} = R'_2 + R'_2 \left(\frac{1-s}{s} \right)$$

Therefore:

$$P_{ag} = 3(I'_2)^2 R'_2 + 3(I'_2)^2 R'_2 \left(\frac{1-s}{s} \right)$$

So,

$$P_{rcl} = 3(I'_2)^2 R'_2 = sP_{ag}$$

$$P_{dev} = 3(I'_2)^2 R'_2 \left(\frac{1-s}{s} \right) = (1-s)P_{ag}$$

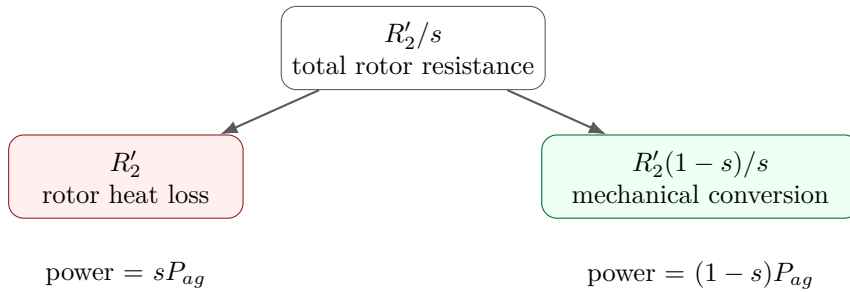


Figure 6: Physical interpretation of R'_2/s . This is the central idea of Phase 5.

Do not mix these three powers

Power	Formula	Meaning
Air-gap power	P_{ag}	Power transferred magnetically from stator to rotor.
Rotor Cu loss	sP_{ag}	Portion wasted as heat in the rotor.
Developed mechanical power	$(1-s)P_{ag}$	Power converted into mechanical form before friction/windage losses.

5 Torque from Power: The Deep Physical Meaning

Mechanical power and torque are related by:

$$P = T\omega.$$

For an induction motor, there are two equally useful views.

Two correct torque relations

$$T_e = \frac{P_{ag}}{\omega_s}$$

This is the electromagnetic torque relation. It uses synchronous angular speed.

$$T_e = \frac{P_{dev}}{\omega_r}$$

This is the mechanical conversion view. It uses rotor speed. Both give the same torque because:

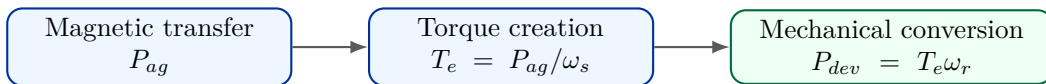
$$P_{dev} = (1 - s)P_{ag}, \quad \omega_r = (1 - s)\omega_s.$$

Proof that both torque expressions are same

$$\frac{P_{dev}}{\omega_r} = \frac{(1 - s)P_{ag}}{(1 - s)\omega_s} = \frac{P_{ag}}{\omega_s}$$

Therefore,

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{P_{dev}}{\omega_r}$$



Torque is born in the air-gap interaction of rotating flux and rotor current. Shaft output comes after mechanical losses.

Figure 7: Torque is connected to air-gap power, not directly to shaft output power.

6 Derivation of the Torque Equation

6.1 Approximate torque equation

Use the simplified series circuit from Section 2.2. Let:

$$R_{eq} = R_1 + \frac{R'_2}{s}, \quad X_{eq} = X_1 + X'_2.$$

Then rotor current referred to stator is:

$$I'_2 = \frac{V}{\sqrt{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2}}.$$

Step-by-step torque equation

Air-gap power:

$$P_{ag} = 3(I'_2)^2 \frac{R'_2}{s}$$

Torque:

$$T_e = \frac{P_{ag}}{\omega_s}$$

Substitute I'_2 :

$$T_e = \frac{3}{\omega_s} \left[\frac{V^2}{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2} \right] \frac{R'_2}{s}$$

Hence,

$$T_e = \frac{3}{\omega_s} \frac{V^2 \left(\frac{R'_2}{s}\right)}{\left(R_1 + \frac{R'_2}{s}\right)^2 + (X_1 + X'_2)^2}$$

Exam language

The torque is produced because the air-gap power crossing from stator to rotor is converted into electromagnetic torque at synchronous angular speed. Therefore, after finding air-gap power from the rotor branch resistance R'_2/s , divide it by ω_s to obtain electromagnetic torque.

6.2 Accurate Thevenin torque equation

For more exact analysis, replace the stator impedance and magnetizing branch by a Thevenin source.

$$T_e = \frac{3}{\omega_s} \frac{V_{th}^2 \left(\frac{R'_2}{s}\right)}{\left(R_{th} + \frac{R'_2}{s}\right)^2 + (X_{th} + X'_2)^2}$$

This is the preferred expression when R_c , X_m , R_1 , and X_1 are not neglected.

Important

Use ω_s , not ω_r , in the air-gap power torque formula. If you use developed mechanical power P_{dev} , then use ω_r . Mixing P_{ag} with ω_r gives a wrong answer.

7 Why Torque is Proportional to Square of Voltage

From the torque equation,

$$T_e = \frac{3}{\omega_s} \frac{V^2 \left(\frac{R'_2}{s} \right)}{\left(R_1 + \frac{R'_2}{s} \right)^2 + (X_1 + X'_2)^2}.$$

For a given slip and fixed machine parameters, the denominator is constant. Hence:

$$T_e \propto V^2$$

Physical interpretation

If voltage is reduced, air-gap flux reduces almost proportionally. Torque depends on interaction of flux and rotor current. Rotor current also depends on induced rotor emf, which is linked to flux. Therefore the combined effect gives approximately square-law behaviour.

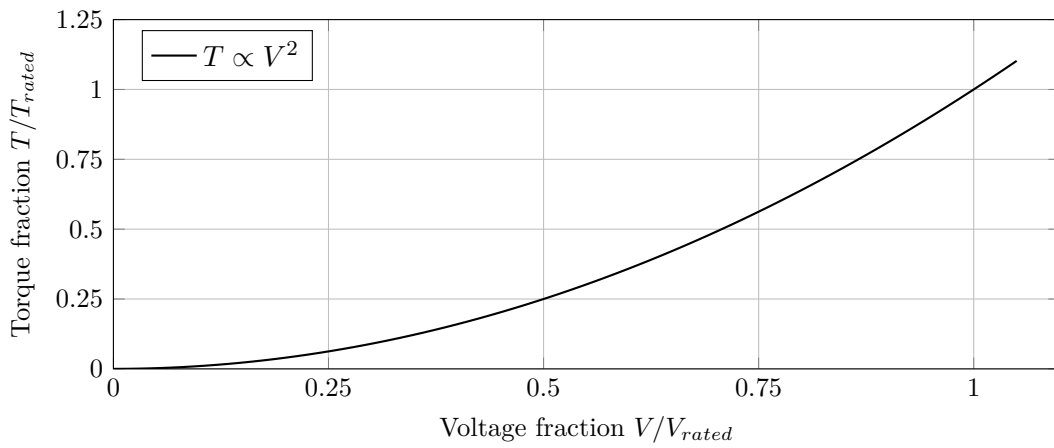


Figure 8: Voltage reduction is dangerous for torque. At 80 percent voltage, torque becomes only 64 percent.

Common numerical shortcut

If voltage is reduced to kV , then torque at the same slip becomes:

$$T_{new} = k^2 T_{old}$$

Example: at $0.8V$, $T_{new} = 0.64T_{old}$.

8 Maximum Torque and Slip at Maximum Torque

For a clean derivation, neglect R_1 or absorb the stator side into Thevenin constants. First use the simpler expression:

$$T_e = K \frac{R'_2/s}{(R'_2/s)^2 + X^2}, \quad K = \frac{3V^2}{\omega_s}, \quad X = X_1 + X'_2.$$

Maximum torque condition

Let:

$$x = \frac{R'_2}{s}$$

Then:

$$T_e = K \frac{x}{x^2 + X^2}$$

Differentiate with respect to x :

$$\frac{d}{dx} \left(\frac{x}{x^2 + X^2} \right) = \frac{(x^2 + X^2) - 2x^2}{(x^2 + X^2)^2} = \frac{X^2 - x^2}{(x^2 + X^2)^2}$$

For maximum torque:

$$X^2 - x^2 = 0 \Rightarrow x = X$$

So:

$$\frac{R'_2}{s_{mT}} = X$$

Therefore,

$$s_{mT} = \frac{R'_2}{X_1 + X'_2}$$

8.1 Maximum torque value

At maximum torque, $R'_2/s = X$. Substitute into torque expression:

$$T_{max} = K \frac{X}{X^2 + X^2} = \frac{K}{2X}$$

Therefore:

$$T_{max} = \frac{3V^2}{2\omega_s(X_1 + X'_2)}$$

Why maximum torque is independent of rotor resistance

Rotor resistance R'_2 decides where maximum torque occurs on the slip axis. But after substituting the maximum torque condition $R'_2/s = X$, R'_2 disappears from the expression of T_{max} . Hence maximum torque is independent of rotor resistance, but the slip at which it occurs is directly proportional to rotor resistance.

8.2 Exact Thevenin maximum torque condition

For the Thevenin circuit,

$$s_{mT} = \frac{R'_2}{\sqrt{R_{th}^2 + (X_{th} + X'_2)^2}}$$

and

$$T_{max} = \frac{3V_{th}^2}{2\omega_s \left[R_{th} + \sqrt{R_{th}^2 + (X_{th} + X'_2)^2} \right]}$$

9 Torque-Slip Characteristics

The torque-slip curve has three major regions.

Region	Slip range	Behaviour
Low slip region	$s \ll s_{mT}$	R'_2/s is large, so torque is approximately proportional to slip. This is normal operating region.
High slip region	$s > s_{mT}$	Reactance dominates and torque decreases as slip increases. This is unstable for motor operation.
Starting point	$s = 1$	Rotor is stationary. Starting current is large and starting torque depends strongly on rotor resistance.

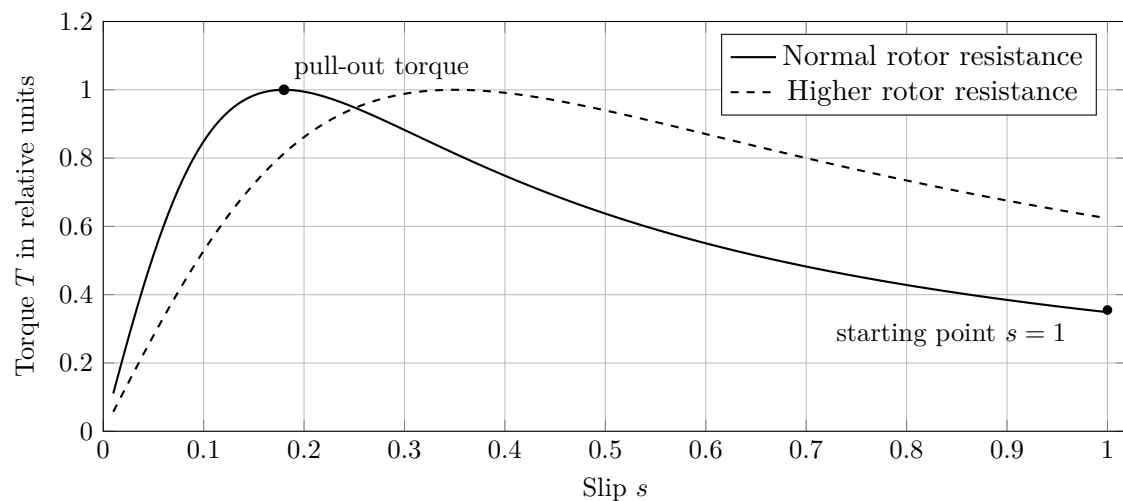


Figure 9: Torque-slip curve. Increasing rotor resistance shifts maximum torque toward higher slip without changing its ideal maximum value.

Why adding rotor resistance helps starting

At starting, $s = 1$. If R'_2 is too small, rotor reactance dominates and starting torque is poor. By increasing rotor resistance, $s_{mT} = R'_2/X$ moves closer to 1. Then maximum torque can be made to occur near starting. This is why slip-ring induction motors can give high starting torque by adding external rotor resistance.

10 Torque-Speed Characteristics

Slip and speed are related by:

$$s = \frac{N_s - N_r}{N_s}, \quad N_r = N_s(1 - s)$$

So, when slip decreases from 1 to 0, rotor speed increases from 0 to N_s .

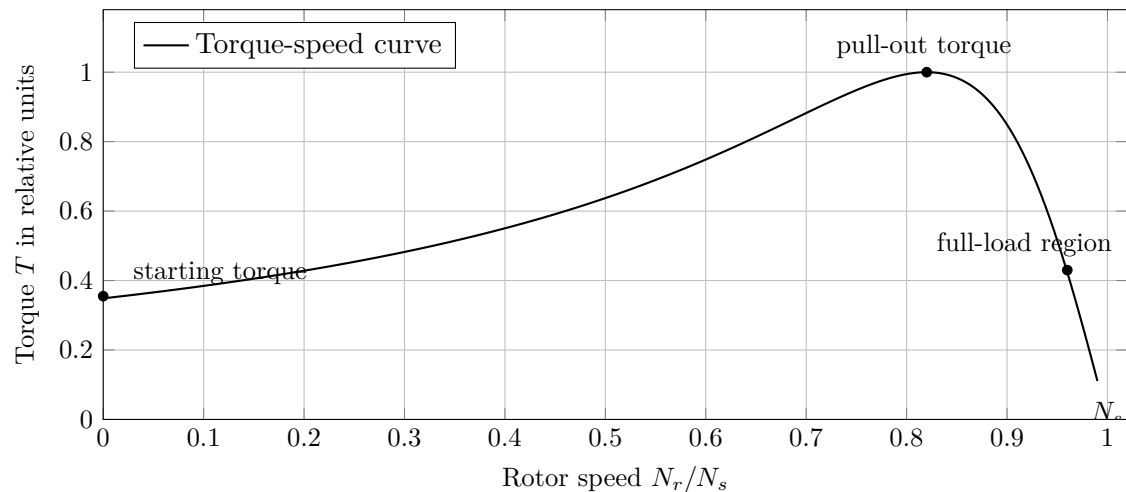


Figure 10: Torque-speed curve. Normal operation is close to synchronous speed, i.e., low slip.

Stability idea

In normal motor operation, if load torque suddenly increases, the rotor slows slightly. Slip increases. In the low-slip region, torque also increases. This extra torque helps the motor regain balance. Hence the low-slip region is stable.

Never draw an induction motor reaching exactly synchronous speed under load

At $N_r = N_s$, slip $s = 0$, rotor emf becomes zero, rotor current becomes zero, and torque becomes zero. A loaded induction motor must run slightly below synchronous speed.

11 Algebra Ratio: T/T_{max}

The ratio form is extremely useful because it removes many constants and directly relates torque to slip.

Derivation of T/T_{max}

Using the simplified expression:

$$T = K \frac{R'_2/s}{(R'_2/s)^2 + X^2}$$

At maximum torque:

$$s_{mT} = \frac{R'_2}{X} \Rightarrow X = \frac{R'_2}{s_{mT}}$$

Substitute $X = R'_2/s_{mT}$:

$$T = K \frac{R'_2/s}{(R'_2/s)^2 + (R'_2/s_{mT})^2}$$

After simplifying and comparing with $T_{max} = K/(2X)$,

$$\frac{T}{T_{max}} = \frac{2}{\frac{s}{s_{mT}} + \frac{s_{mT}}{s}}$$

How to use this in exam

If a question gives full-load slip and maximum-torque slip, use:

$$\frac{T_{FL}}{T_{max}} = \frac{2}{\frac{s_{FL}}{s_{mT}} + \frac{s_{mT}}{s_{FL}}}$$

This avoids calculating current and impedance separately.

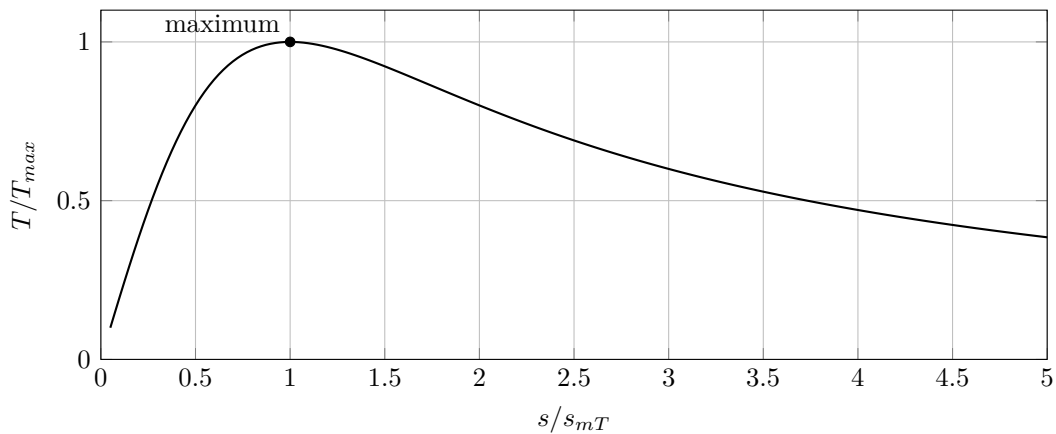


Figure 11: Universal torque ratio curve. Maximum occurs at $s/s_{mT} = 1$.

12 Rotor Current Ratios

The referred rotor current in the simplified circuit is:

$$I'_2(s) = \frac{V}{\sqrt{\left(\frac{R'_2}{s}\right)^2 + X^2}}$$

where $X = X_1 + X'_2$ if stator resistance is neglected.

Starting current to running rotor current ratio

At starting, $s = 1$:

$$I'_{2st} = \frac{V}{\sqrt{R_2'^2 + X^2}}$$

At any slip s :

$$I'_2 = \frac{V}{\sqrt{(R'_2/s)^2 + X^2}}$$

Therefore,

$$\boxed{\frac{I'_{2st}}{I'_2} = \frac{\sqrt{(R'_2/s)^2 + X^2}}{\sqrt{R_2'^2 + X^2}}}$$

Maximum-torque current to running current ratio

At maximum torque, $s = s_{mT}$, so $R'_2/s_{mT} = X$:

$$I'_{2mT} = \frac{V}{\sqrt{X^2 + X^2}} = \frac{V}{\sqrt{2}X}$$

Thus,

$$\boxed{\frac{I'_{2mT}}{I'_2} = \frac{\sqrt{(R'_2/s)^2 + X^2}}{\sqrt{2}X}}$$

Do not confuse two rotor-current formulas

Actual rotor-frequency form:

$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}}$$

Slip-corrected referred circuit form:

$$I'_2 = \frac{E'_2}{\sqrt{(R'_2/s)^2 + (X'_2)^2}}$$

Both are consistent, but do not mix pieces of one formula with the other.

13 Maximum Mechanical Power Slip

Maximum torque and maximum mechanical power do not generally occur at the same slip. This is a common advanced exam trap.

For the simplified model:

$$P_{dev} = 3(I'_2)^2 R'_2 \left(\frac{1-s}{s} \right)$$

Using $I'_2 = V / \sqrt{(R'_2/s)^2 + X^2}$, we get:

$$P_{dev} = 3V^2 \frac{R'_2 s(1-s)}{R'^2_2 + X^2 s^2}$$

Slip for maximum mechanical power

Let constants be ignored and maximize:

$$f(s) = \frac{s(1-s)}{R'^2_2 + X^2 s^2}$$

Differentiate and set derivative equal to zero:

$$\frac{d}{ds} \left(\frac{s - s^2}{R'^2_2 + X^2 s^2} \right) = 0$$

This gives:

$$R'^2_2 - 2R'_2 s - X^2 s^2 = 0$$

or,

$$X^2 s^2 + 2R'_2 s - R'^2_2 = 0$$

Solving the positive root:

$$s_{mP} = \frac{R'_2}{\sqrt{R'^2_2 + X^2} + R'_2}$$

Using $s_{mT} = R'_2/X$:

$$s_{mP} = \frac{s_{mT}}{\sqrt{1 + s^2_{mT}} + s_{mT}}$$

Exam understanding

Torque is P_{ag}/ω_s . Mechanical power is $T\omega_r$. Near standstill, speed is low, so mechanical power can be low even if torque is large. Therefore maximum mechanical power occurs at a lower slip than maximum torque in the usual motor region.

14 Efficiency and Performance Quantities

14.1 Efficiency

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

where

$$P_{out} = P_{dev} - P_{rotational\ losses}.$$

14.2 Speed regulation

For an induction motor, speed regulation is usually:

$$\%SR = \frac{N_{nl} - N_{fl}}{N_{fl}} \times 100\%$$

Since no-load speed is close to synchronous speed:

$$\%SR \approx \frac{N_s - N_{fl}}{N_{fl}} \times 100\%.$$

14.3 Performance table

Quantity	Formula	How to interpret
Slip	$s = (N_s - N_r)/N_s$	Measures how much the rotor lags behind the rotating field.
Rotor frequency	$f_2 = sf_1$	At starting, $f_2 = f_1$; at near synchronous speed, f_2 is small.
Air-gap power	$P_{ag} = 3(I_2')^2 R_2'/s$	Total power transferred across the air gap.
Rotor Cu loss	$P_{rcl} = sP_{ag}$	Rotor heat loss.
Developed power	$P_{dev} = (1 - s)P_{ag}$	Mechanical power before rotational losses.
Torque	$T_e = P_{ag}/\omega_s$	Electromagnetic torque.
Output power	$P_{out} = P_{dev} - P_{rot}$	Useful shaft power.
Efficiency	P_{out}/P_{in}	Fraction of input converted into useful output.

15 Worked Numerical: Full Power-Flow Method

Problem

A 3-phase induction motor takes $P_{in} = 15 \text{ kW}$. Stator copper loss is 600 W, core loss is 400 W, and rotor copper loss is 700 W. Rotational losses are 500 W. Find air-gap power, slip, developed mechanical power, shaft output, and efficiency.

Solution

Input power:

$$P_{in} = 15000 \text{ W}$$

Air-gap power:

$$P_{ag} = P_{in} - P_{scl} - P_c = 15000 - 600 - 400 = 14000 \text{ W}$$

Slip:

$$s = \frac{P_{rcl}}{P_{ag}} = \frac{700}{14000} = 0.05$$

So slip is 5%.

Developed mechanical power:

$$P_{dev} = (1 - s)P_{ag} = 0.95 \times 14000 = 13300 \text{ W}$$

Shaft output:

$$P_{out} = P_{dev} - P_{rot} = 13300 - 500 = 12800 \text{ W}$$

Efficiency:

$$\eta = \frac{P_{out}}{P_{in}} \times 100 = \frac{12800}{15000} \times 100 = 85.33\%$$

How to think

You did not need a complicated circuit for this numerical. Once losses are given, just move along the power-flow diagram. This is why the power-flow diagram is the first diagram to draw in exam solutions.

16 Worked Numerical: Maximum Torque Ratio

Problem

A motor has maximum torque at $s_{mT} = 0.20$. Find the ratio of full-load torque to maximum torque if full-load slip is $s = 0.04$.

Solution

Use:

$$\frac{T}{T_{max}} = \frac{2}{\frac{s}{s_{mT}} + \frac{s_{mT}}{s}}$$

Substitute:

$$\begin{aligned} \frac{T_{FL}}{T_{max}} &= \frac{2}{\frac{0.04}{0.20} + \frac{0.20}{0.04}} \\ &= \frac{2}{0.2 + 5} = \frac{2}{5.2} = 0.3846 \end{aligned}$$

So,

$$\boxed{T_{FL} = 0.3846 T_{max}}$$

Meaning

At full load the motor usually operates far to the low-slip side of maximum torque. Maximum torque is not the normal operating torque; it is the pull-out limit.

17 Graphical Summary: What Changes What?

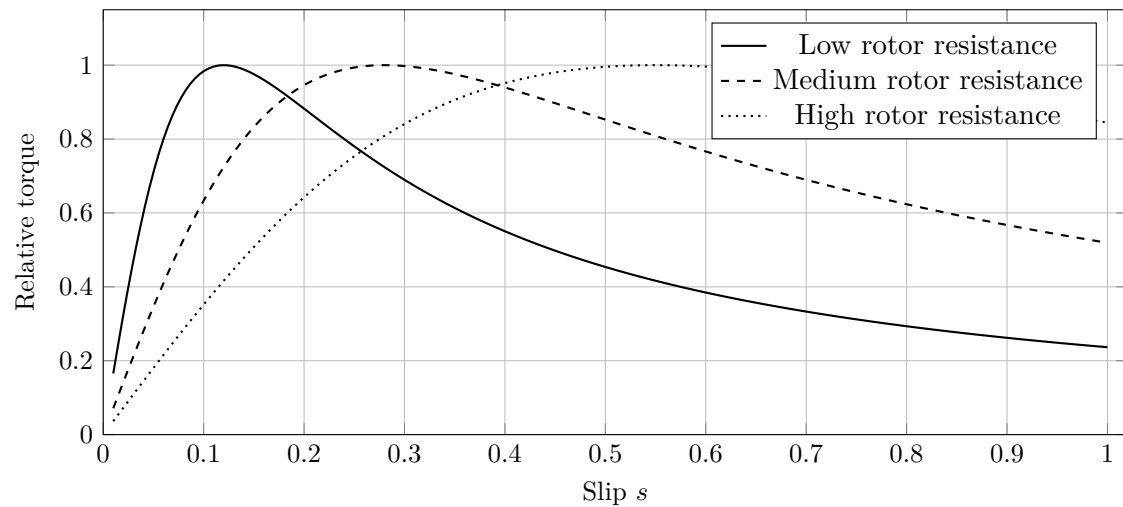


Figure 12: Rotor resistance shifts the curve horizontally toward higher slip. Ideal maximum torque stays same if voltage and reactance are unchanged.

Parameter changed	Result
Supply voltage V decreases	Whole torque curve drops approximately as V^2 .
Rotor resistance R'_2 increases	Slip at maximum torque increases. Starting torque improves up to an optimum.
Leakage reactance $X_1 + X'_2$ increases	Maximum torque decreases and maximum torque slip decreases for fixed R'_2 .
Load torque increases	Motor slows slightly, slip increases, and electromagnetic torque rises in the stable region.

18 Exam-Oriented Derivation Templates

Template 1: Derive torque equation

1. Draw the approximate equivalent circuit with R_1 , X_1 , R_2'/s , and X_2' in a correctly closed series path.

2. Write rotor current:

$$I_2' = \frac{V}{\sqrt{(R_1 + R_2'/s)^2 + (X_1 + X_2')^2}}$$

3. Write air-gap power:

$$P_{ag} = 3(I_2')^2 R_2'/s$$

4. Write torque:

$$T_e = P_{ag}/\omega_s$$

5. Substitute and box the final result.

Template 2: Prove maximum torque independent of rotor resistance

1. Start from $T = K \frac{R_2'/s}{(R_2'/s)^2 + X^2}$.

2. Put $x = R_2'/s$.

3. Differentiate $x/(x^2 + X^2)$.

4. Get condition $x = X$, so $s_{mT} = R_2'/X$.

5. Substitute $x = X$ into torque and show $T_{max} = K/(2X)$, which has no R_2' .

Template 3: Draw torque-speed curve

Draw speed on x-axis from 0 to N_s . Mark starting torque at $N_r = 0$, pull-out torque before synchronous speed, full-load point close to N_s , and zero torque at N_s . Mention: normal operation is low-slip and stable.

19 Final Revision Sheet

$$s = \frac{N_s - N_r}{N_s},$$

$$N_s = \frac{120f}{P},$$

$$\omega_s = \frac{2\pi N_s}{60}$$

$$P_{ag} = 3(I_2')^2 \frac{R_2'}{s},$$

$$P_{rcl} = sP_{ag},$$

$$P_{dev} = (1 - s)P_{ag}$$

$$T_e = \frac{P_{ag}}{\omega_s} = \frac{P_{dev}}{\omega_r}$$

$$T_e = \frac{3}{\omega_s} \frac{V^2(R_2'/s)}{(R_1 + R_2'/s)^2 + (X_1 + X_2')^2}$$

$$T_e \propto V^2$$

$$s_{mT} = \frac{R_2'}{X_1 + X_2'} \quad (\text{approx.})$$

$$\frac{T}{T_{max}} = \frac{2}{\frac{s}{s_{mT}} + \frac{s_{mT}}{s}}$$

$$s_{mP} = \frac{R_2'}{\sqrt{R_2'^2 + X^2 + R_2'}}$$

$$\eta = \frac{P_{out}}{P_{in}} \times 100\%$$

Last mental image

The circuit gives current. Current through R_2'/s gives air-gap power. Air-gap power divided by synchronous speed gives torque. Slip decides how air-gap power divides into rotor heating and mechanical conversion. This one chain solves most induction motor performance questions.